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Quality Assessment of Stereophotoclinometry as a Shape Modeling Method Using a Synthetic Asteroid

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Abstract

The stereophotoclinometry (SPC) software suite has been used to generate global digital terrain models (DTMs) of many asteroids and moons, and was the primary tool used by the Origins, Spectral Interpretation, Resource Identification, and Security-Regolith Explorer (OSIRIS-REx) mission to model the shape of asteroid Bennu. We describe the dedicated preflight testing of SPC for the OSIRIS-REx mission using a synthetic “truth” asteroid model. SPC has metrics that determine the internal consistency of a DTM, but it was not known how these metrics are related to the absolute accuracy of a DTM, which was important for the operational needs of the mission. The absolute accuracy of an SPC-generated DTM cannot be determined without knowing the truth topography. Consequently, we developed a realistic, but synthetic, computer-generated representation of asteroid Bennu, photographed this synthetic truth model in an imaging campaign similar to that planned for the OSIRIS-REx mission, and then generated a global SPC DTM from these images. We compared the SPC DTM, which was represented by a radius every 70 cm across the asteroid surface, to the synthetic truth model to assess the absolute accuracy. We found that the internal consistency can be used to determine the 3D root-mean-square accuracy of the model to within a factor of two of the absolute accuracy.

Unified Astronomy Thesaurus concepts: [Near-Earth objects \(1092\)](#); [Planetary science \(1255\)](#)

1. Introduction

The primary goal of the Origins, Spectral Interpretation, Resource Identification, and Security-Regolith Explorer (OSIRIS-REx) mission is to return at least 60 g of regolith from asteroid (101955) Bennu to Earth (Lauretta et al. 2017). The spacecraft collected the sample in 2020 October and departed Bennu for Earth in 2021 May (Lauretta et al. 2021). To achieve this goal, a global digital terrain model (DTM) of the asteroid was needed for navigating the spacecraft around and down to the surface of Bennu. The OSIRIS-REx mission used the stereophotoclinometry (SPC) software suite to construct this global DTM of Bennu. SPC, as the name suggests, uses a photoclinometric solution that is constrained by a stereo solution. Photoclinometry determines surface tilt and albedo, while stereo determines positions in 3D space. Combining the two techniques enables a solution that overcomes the limitations of both techniques (Gaskell et al. 2008; Palmer et al. 2022).

Although SPC has been used to construct DTMs of at least 17 solar system objects, including asteroids, comets, and several terrestrial planets (see Palmer et al. 2022 for a full list), a systematic evaluation of its accuracy relative to a ground truth has been lacking. At best, comparisons could be made with other DTM generation techniques that also lacked ground truth

(see Barnouin et al. 2020 and references therein). One previously published test (Craft et al. 2020) used a physical object for a real-world validation of SPC that removed the systematic artifacts that come from fully digital experiments. Results indicated that the SPC-derived DTM matched the surface shape measured by a high-precision laser altimeter, with root-mean-square (rms) values of less than a few camera pixels. However, this previous test did not include realistic spacecraft orbit and attitude conditions, with their associated uncertainties, and it was limited to a local DTM.

This paper presents results from the ground-based testing performed in 2016, prior to the spacecraft encounter with Bennu, that used a synthetic, computer-generated representation of a Bennu-like asteroid to evaluate the accuracy of a global SPC-derived DTM. This approach allowed control and assessment of all aspects of the image position and pointing, including many of the orbit and pointing uncertainties typically associated with a spacecraft mission. This evaluation was conducted in support of the OSIRIS-REx proximity operations at asteroid Bennu, with the purpose of understanding the fidelity of the model for scientific and navigation purposes. Because the SPC software suite was part of the operational navigation system, it was required to meet NASA’s Class B software standards (NASA 2014). This study provided one of the verification tests that demonstrated that SPC could meet the mission requirements. Here we show how we assessed the expected accuracy of the global SPC-derived DTM of Bennu, and discuss implications for future missions.



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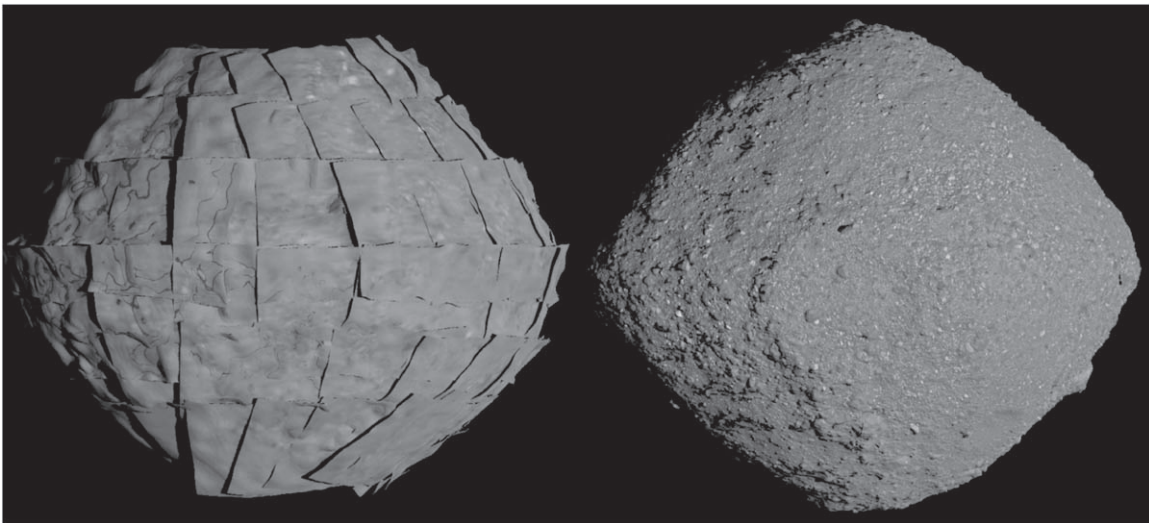


Figure 1. The left panel shows the initial stages of a maplet-based SPC model before the topographic integration phase. The initial maplets poorly represent the topography. The right panel shows the final shape after the maplets have seamlessly flexed into the proper shape.

The synthetic asteroid, or “truth” model, was based on the premission radar model of Bennu (Nolan et al. 2013a, 2013b) with superposed simulated topography (Barnouin et al. 2020). We used SPC to construct a DTM from simulated images of the truth model, with no prior knowledge of the latter’s shape. Thus, only internal SPC metrics (that will be discussed in Palmer et al. 2022) were used to assess which areas of the SPC model needed further refinement. Various versions of the SPC model were archived to track its progression. Only once construction of the SPC model was complete were it and its intermediate stages compared to the truth model.

In the following sections, we detail how the synthetic truth model was constructed, the campaign for imaging the truth model and how it differed from that flown at Bennu, how we generated test images, and how we generated the SPC model using these images. Comparisons of geometric differences between the SPC model and the truth model provide the basis for the assessment of the quality of the SPC product. Normally, the SPC metrics discussed in Palmer et al. (2022) are based on assessments of the internal consistency of the SPC model with the data used to construct it, and can be determined in flight. Thus, we can establish the relationship between the absolute errors of the SPC model (something we cannot determine in flight) and the uncertainty we predict based on the internal SPC metrics that are produced during the SPC model construction. Details of additional testing, and how error propagates into the SPC model, will be discussed in E. E. Palmer et al. (2022b, in preparation).

2. Methods

2.1. Overview

All results that we present here are from the ground-based SPC testing performed in 2016, prior to the launch of the OSIRIS-REx spacecraft. Because these tests used the mission as a template, much of the terminology is similar. Nonetheless, there are differences between these tests and the mission (see Section 2.5).

The OSIRIS-REx mission required two different global SPC models for proximity operations. The first model, known as the 75 cm SPC model, was required by the mission to switch from

stellar navigation to optical surface navigation, a task that is necessary for most planned observations. This 75 cm SPC model was created from images captured during the initial Approach and Preliminary Survey phases (Lauretta et al. 2021); in the preflight SPC tests, the best simulated image resolutions for this period were about 12 cm pixel^{-1} . The second model, known as the 35 cm SPC model, was required to obtain the higher-resolution topography needed to select a location from which to collect a sample. This 35 cm model was created mostly from Preliminary Survey and Baseball Diamond imaging (Lauretta et al. 2021), though some images from the last few days of Approach were also used. In the preflight SPC tests, the resolution of the simulated images incorporated into the 35 cm SPC model reached down to about 5 cm pixel^{-1} . The 75 and 35 cm SPC models are named after the ground sample distances (GSDs) of the local-scale maplets (see Section 2.2) that are used to build them, and their names do not represent the accuracy or GSD of the global model. Global SPC models all have 1.6 million vectors to represent the object, so larger objects will have radius values with larger separation. Both global SPC models represent a radius value roughly every 70 cm for an object the size of Bennu.

2.2. How SPC Works

SPC blends the benefits of the 3D absolute geometric calculation provided by stereo photogrammetry with the high-resolution vertex-to-vertex slope calculations provided by 2D multi-image photoclinometry. SPC covers the surface with overlapping DTMs (called maplets) that are 99×99 vertices at a given GSD. Every image that contains the maplet’s terrain is assigned and aligned to that maplet. The position of these maplets in 3D space is calculated by stereo. The interstitial elevations are calculated by integrating the x -slope and y -slope generated by photoclinometry. Corrections are calculated by the difference in data number between each image and its corresponding rendered maplet. Multiple iterations improve the DTM to the point where the overlapping regions fuse together, making a coherent final 3D object (see Figure 1). Another key aspect of SPC is the ability to determine the spacecraft position and pointing for each image based upon its assigned maplets.

Further details about the practical application of SPC can be found in Palmer et al. (2022 and references therein).

2.3. Test Conditions

The suite of tests was designed to simulate the proximity operation activities when the spacecraft arrived at Bennu. This required simulating a synthetic truth model with realistic terrain and navigation flight profiles with realistic flight and imaging uncertainties. Previous tests enabled the generation of these components to be streamlined, but the generation of fully consistent test data was a significantly larger effort than the testing itself. Not only did the data have to be generated, but then they had to be packaged in the same formats that would be used by the mission. Examples include images as FITS files with correct headers, and navigation trajectories in the NAIF SPICE kernel format (Acton et al. 2018).

One simplification that was made for these tests was to start without errors in the planetary data constants for Bennu, i.e., the starting model parameters for the R.A. and decl. of the pole and the rotation rate were the same as in the truth model. Subsequent tests iterating between the mission's Altimetry Working Group (AltWG) and navigation team focused on those parameters. These efforts will be described in C. D. Adam et al. (2022, in preparation).

Two sets of kernels were generated for these tests. The baseline flight trajectories were generated by the navigation team. These flight trajectories were used to generate the spacecraft position kernels. Pointing kernels were then generated to simulate the observations described in the OSIRIS-REx Design Reference Mission. For testing purposes, a second set of kernels was generated that included spacecraft knowledge errors. Thus, when the test was run, the kernels with flight-realistic errors were provided to the AltWG to simulate the reconstructed kernels that would be provided by the navigation team during flight. The 1σ flight errors in the kernels were 400 m during Approach, 100 m during Preliminary Survey, and 6.4 m during Detailed Survey.

2.4. Generation of the Synthetic Asteroid (Truth Model)

The synthetic truth model used for these tests was the fourth and final DTM of the Bennu simulated preflight by AltWG for the OSIRIS-REx mission (Barnouin et al. 2020). As a starting point, AltWG made use of the radar model of Bennu generated by Nolan et al. (2013a), and modified it using the CreatorP software (Hergenrother et al. 2014; Seabrook et al. 2019). E. E. Palmer et al. (2022b, in preparation) describe in detail how the AltWG used CreatorP to produce the simulated surface topography. This synthetic truth model and the SPC models produced by this work can be found in Palmer & Gaskell (2021).

The expected realistic terrains (see Barnouin et al. 2020 for rationale) that were added to the radar DTM of Bennu included boulders, craters, and variations in roughness. A sinusoidal dependence with longitude was applied to the boulder and crater density, so that these would be small at 0° and high at 180° longitude. Rocks 5 cm to 10 m in diameter had a size power law of $10,000^{-2.7}$ rocks km^{-2} , while rocks $10\ \mu\text{m}$ to 5 cm in diameter had a size power law of $10,000^{-2.9}$ rocks km^{-2} . The rocks had a random ellipsoidal footprint with a random orientation. Craters also had a size power law of $10,000^{-2.9}$ with no ejecta blanket. Like the boulder and crater

density, surface roughness was made to vary sinusoidally, with a maximum roughness at 90° and 270° and a minimum roughness at 0° and 180° longitude. Figure 2 shows the truth shape and SPC model. Figures 3 and 4 (top) show the albedo and radius of the truth shape, respectively.

Most of this truth model had surfaces with slopes, craters, and boulders that would make it unsafe for the spacecraft to sample. A 50 m diameter simulated sampling site was located in a smooth and relatively boulder- and crater-free region close to the equator at -10° latitude and 350° east longitude (no such large hazard-free region was identified by the mission on Bennu itself; see Lauretta et al. 2021; Norman et al. 2022; Olds et al. 2022 in this focus issue). The truth shape needed a resolution capable of supporting the highest-resolution images, so it was given a global GSD of 5 cm and a sampling site GSD of 1 cm.

2.5. Cameras and Imaging Campaign

The relevant instruments for these simulations were PolyCam and MapCam, which are part of the OSIRIS-REx Camera Suite (OCAMS) (Rizk et al. 2018). These instruments were simulated using the parameters in E. E. Palmer et al. (2022b, in preparation). All simulated images were 1024 by 1024 pixels and had no point-spread function, geometric distortion, or radiometric calibration parameters applied. Based upon theoretical knowledge and previous mission experience, we were (and still are) confident that we could correct for these camera effects without introducing significant error into the images. Our tests on a physical object with real cameras also showed that we could remove these effects and still produce a high-quality DTM (Craft et al. 2020). The simulated images used to build the SPC model corresponded to the plans for the early phases of the OSIRIS-REx proximity operations. These phases included Approach, Preliminary Survey, and the Baseball Diamond portion of Detailed Survey (Gal-Edd & Chevront 2015). Baseball Diamond provided the highest-resolution images and, at the time of these tests, consisted of two arcs over both the northern and southern hemisphere of the truth model to obtain complete coverage. Each of the four arcs contained nadir and off-nadir images that provided a small stereo angle of $\sim 15^\circ$. See Lauretta et al. (2017, 2021) for details of the OSIRIS-REx mission phases as planned and flown, respectively, which in some cases differed dramatically. The mission was designed such that the ranges of phase angles (Sun to surface to spacecraft angle), emission angles (vertical to spacecraft angle), and incidence angles (vertical to Sun angle) in the collected images would provide a robust data set to generate a 35 cm SPC model (Weirich et al. 2017; Palmer et al. 2018, 2022; Antreasian et al. 2019; Barnouin et al. 2020; Wibben et al. 2020; Al Asad et al. 2021).

During the simulated Approach, the spacecraft collected increasingly close-range images of the truth model over the equator on the lit side. The simulated images were taken between about 100 and 30 km from the synthetic asteroid by PolyCam. The first five days of simulated imaging were of a full rotation of a full-frame truth model, with the truth model rotating 10° between each image. The next four days were also of a single rotation, but with a 2×2 image grid of the truth model, with 10° of rotation between each grid. An image grid was used by the mission at these (relatively) close distances because the spacecraft position and pointing error was still large enough that a single image may not have captured Bennu.

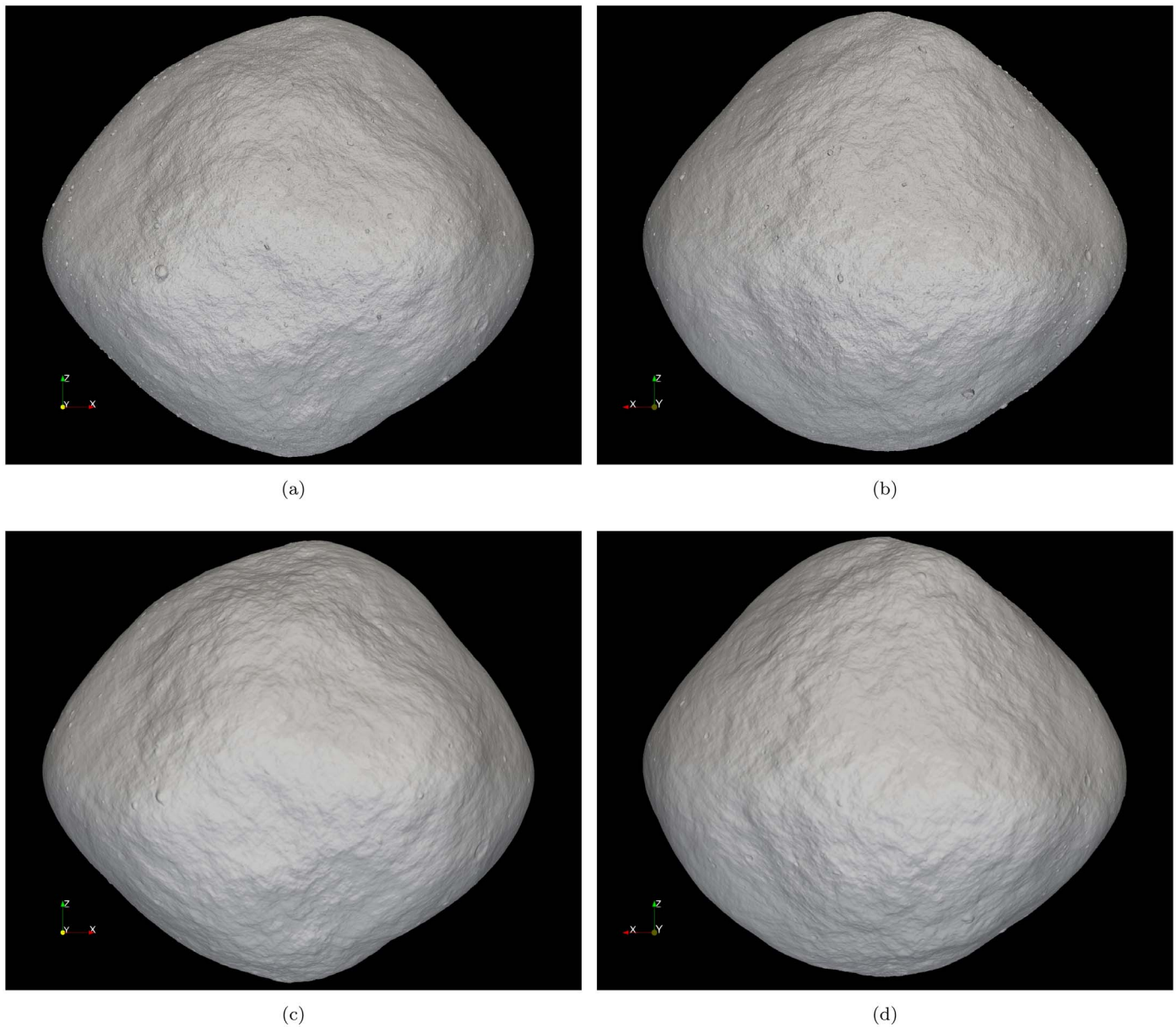


Figure 2. The computer-generated synthetic asteroid (a.k.a. truth model) used for these tests is shown in (a) and (b), and the SPC model is shown in (c) and (d), both rendered with artificial lighting for clarity. The $+y$ -axis points into the page in panels (a) and (c), and out of the page in panels (b) and (d). The overall shape of the SPC 35 cm model closely matches the truth model. Minor differences in the boulders, especially shape and height, can be seen. Additional processing of the boulders could have been performed, but was outside the scope of these tests.

For the last two days of simulated imaging, the spacecraft drifted north in anticipation of the upcoming north pole pass of Preliminary Survey.

The simulated Preliminary Survey phase consisted of one imaging pass each over the north pole, the equator, and lastly the south pole of the synthetic asteroid, at ranges of between 7 and 8 km. Most simulated Preliminary Survey data were collected by PolyCam, but for the north and south pole passes, MapCam was used for a time near the closest approach. MapCam acquired full-frame images of the north and south poles to constrain the radius at the equator. The north polar pass began east and ended west of the asteroid. The equatorial pass was west to east over the equator. The south polar pass began west and ended east of the synthetic asteroid.

The simulated Baseball Diamond campaign consisted of four observing stations for the northern hemisphere and four

observing stations for the southern hemisphere, all of which were at midlatitudes. Further details of the simulated images from all of the mission phases considered here, such as the number and maximum resolution in each phase, are presented in Table 1.

As alluded to above, we executed these tests using an early version of the plans for the imaging campaign. The lessons learned allowed the mission to respond to unexpected operational limits and modify the imaging campaign before the actual spacecraft arrived at Bennu. Preliminary Survey in the simulations had a higher image resolution than that actually flown, but Baseball Diamond in the simulations had fewer unique incidence and emission angles. Ultimately, the imaging campaign as flown (Lauretta et al. 2021; Barnouin et al. 2022) was an improvement for SPC relative to the conditions we tested.

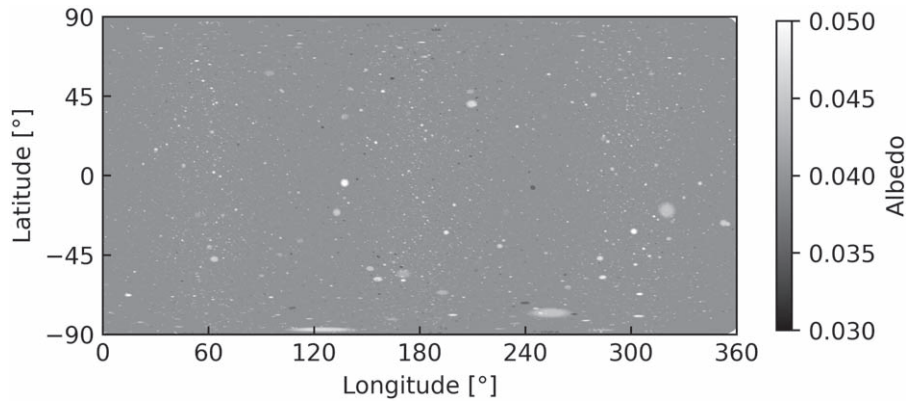


Figure 3. Albedo of the truth model, plotted as a simple cylindrical projection and scaled to an average albedo of 4%.

2.6. Generation of Test Images

Simulated spacecraft trajectories (see Section 2.5) and camera parameters were used to create the OCAMS-like simulated images of the truth shape. These images were generated using the Geomod geometry modeling tool and the Phillum physically based stochastic ray tracer, both developed at Goddard Space Flight Center as part of the Freespace simulation environment. The photometric model used to generate the images is the same Lunar-Lambert function used by SPC (McEwen 1991). This produced photorealistic synthetic images for which we knew the truth relative positions.

The generation of the images for these tests closely follows what would be acquired from a real spacecraft; however, there are some differences. To represent a realistic flight-like profile, the spacecraft would need to be perturbed from the nominal trajectory, and then the image would be generated. SPC would then attempt to accurately adjust the spacecraft position from the nominal position to the perturbed position. Generating the images with a flight-like profile increases the image generation complexity. Because this increased complexity would not test the ability of SPC to create an accurate DTM, the flight-like profile was not used. For these tests, the images were instead generated using a nominal flight profile, and then assigned to a kernel with a flight-like profile.

The perturbation from the nominal spacecraft position was based on the position uncertainty expected from the mission design. The mission design involves a 1σ uncertainty distribution, which means $\sim 70\%$ of the deviations are within that distance. This 1σ position uncertainty varied between and within the simulated mission phases, but was generally hundreds of meters during simulated Approach, 20–100 m during simulated Preliminary Survey, and about 5 m during simulated Baseball Diamond. If the distance of the perturbation of an image was greater than 1σ of the position uncertainty, the perturbation was set to the 1σ value. This allowed for a simple image generation process that still tested the ability of SPC to find the position of the spacecraft when the image was acquired.

It is worth noting a peculiarity of the image grids taken during Approach. Due to the error in the position and pointing of the spacecraft, the spacecraft had to take 2×2 , 3×3 , and 4×5 image grids to ensure a 3σ probability that Bennu would be in one of those images. Bennu did not fill the field of view until the last two days of simulated Approach. For these tests, the centers of the simulated image grids were targeted to be the

center of the truth model. During the actual operations, the spacecraft’s position and pointing errors made it such that Bennu could be contained in one, two, or four of the grid images. However, in these tests, four simulated images from the grid were always required, greatly increasing processing time. This paradoxically means that in a real-world scenario, the image grids would be faster to process than in the test scenario.

2.7. Generation of the SPC-derived Global DTM

Table 2 details the steps of processing to generate the SPC models from the simulated images during the preflight testing. Between each step in the table, a topography iteration was performed (Palmer et al. 2022). Detailed descriptions of these steps are provided in E. E. Palmer et al. (2022b, in preparation). “Approach Day 1” images are not the first images we would acquire during the approach to the real Bennu, but rather are the first day of images suitable for SPC. All simulated images from Approach Day 1 were aligned to the Nolan et al. (2013a) radar-based truth model as a starting point. These images were then used to build 120 cm GSD maplets, with the location of each maplet defined by the latitude, longitude, and starting topography of each maplet loaded from the Nolan et al. (2013a) shape. The maplets were then iterated (Palmer et al. 2022) to begin warping them to the heights indicated by the images. The 120 cm maplets were then used to generate a new shape.

We then ingested Approach Day 2 simulated images, aligned them to the updated shape, added them to the maplets, and performed an iteration (Palmer et al. 2022) to generate a new shape again. After ingesting simulated images from Approach Day 5, we performed an internal error check to identify and fix any errors, such as misaligned images or topography, that needed additional processing. As part of this process, we also performed the SPC version of a bundle adjustment, which is updating the spacecraft position and pointing, and then generated a new SPC model. Next, we generated 75 cm GSD maplets where the base topography was loaded from the most recent SPC model.

Then we ingested the simulated Approach images from Day 6 through 9. Beginning with these gridded Approach images, and continuing throughout the rest of the simulated imaging campaigns, some of the simulated images were not usable by SPC and were therefore excluded. The synthetic asteroid was sometimes not in the frame of the camera, and other times the

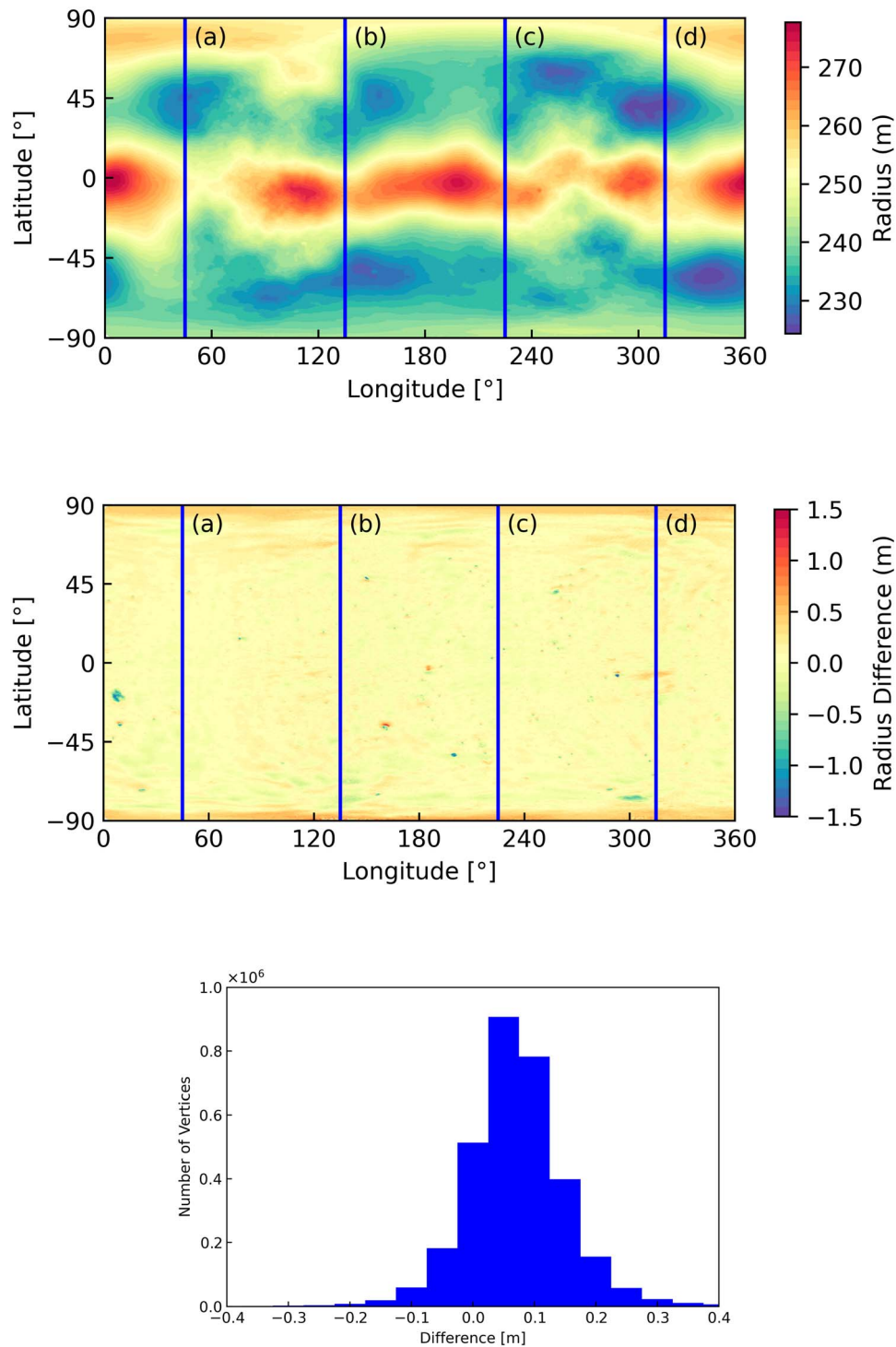


Figure 4. Top: the local radius from the truth model as a simple cylindrical projection. Sectors (a) and (c) represent rough truth model terrain with a medium density of craters and boulders, (b) represents smooth truth model terrain with a high density of craters and boulders, and (d) represents smooth truth model terrain with a low density of craters and boulders. Middle: the difference in radius from the CoF between the SPC model and the truth model, with the CoF drift removed. The maximum difference is 1.5 m, and in the majority of locations is less. Bottom: the radius difference from the middle panel displayed as a histogram.

simulated images captured the limb, resulting in a high emission angle (i.e., $>70^\circ$).

After incorporating the images from the simulated Preliminary Survey passes, we generated 35 cm GSD maplets, again with the topography loaded from the newly updated shape. After a final round of error checking and updating, the 75 cm SPC model was complete. Note that for the 75 cm shape, we were most interested in the 75 cm maplets, as they were tied

to navigation requirements and fed into subsequent testing. However, the 35 cm maplets improved the alignment of the images and the quality of the 75 cm maplets because information is shared between maplets.

We then ingested the images from the four simulated Baseball Diamond passes and generated 18 cm GSD maplets. For the 18 cm maplets, loading topography from the most recent global SPC model would only provide an effective GSD

Table 1
Summary of Simulated Images

Mission Phase	Range (km)	Number of Images	Phase Angle (deg)	Best GSD (cm pixel ⁻¹)	Landmarks per Image
Approach	140.5–30.0	927	4–41	41	223
Preliminary Survey	11.1–8.9	877	37–92	12	187
Detailed Survey	3.8	3265	44–52	5	54

Note. Details of the simulated images and the parameters used to create them. Best GSD provides the best value for the central pixel in the simulated images. Landmarks per image provides the mean number of stereo control points in each simulated image, which is equivalent to the number of maplets that use the image. Images were synthesized using the NASA Goddard Freespace simulation tool using pixel fields of view provided by Rizk et al. (2018).

of about 70 cm, which is not high enough to properly align the 5 cm images. Instead we created large regional DTMs with a GSD of 22 cm from the 35 cm maplets. These regional DTMs, when taken together, fully covered the global shape. These regional DTMs were then used to define the location where each maplet would be generated, and the initial topography of the maplet was loaded from these regional DTMs. After a final round of error checking, the 35 cm shape was complete. Similar to the case of the 75 cm shape, the mission was most interested in the 35 cm maplets for navigation requirements, but the 18 cm maplets improved the 35 cm maplets.

When constructing a global DTM with SPC or other software suites, an important consideration is the potential for a systematic error in the DTM's overall size. This is where the radius of the DTM is slightly too big or too small. SPC has various techniques to identify and correct for size bias, one of which was developed after these tests and was used during the proximity operations at Bennu. However, these techniques were not needed during this SPC DTM generation because the errors used in the test were small enough to avoid problems.

The level of effort for the various SPC tasks was recorded during the SPC model generation. This was necessary for operational planning because it determined the fastest possible cadence for mission stages and resources. We broke down the SPC tasks into interactive processing (i.e., human with a keyboard) and batch processing (i.e., computer processing without input from a human) tasks. The level of effort breakdown is shown in Table 2. The total processing (i.e., interactive and batch processing) for the 75 cm SPC model took 393 hr. However, that rate of progress could only be achieved by multiple people working in shifts. A single person working 40 hr a week would have required 38 calendar days to generate the 75 cm model. For the 35 cm model, total processing took 679 hr, or 67 calendar days for a single person. An interesting point is that the ratio between calendar days and processing days was about 2.3 for both the 35 and 75 cm SPC models. However, 33% of the time needed to generate the 75 cm SPC model was spent in interactive processing, whereas only 7% of the time needed to generate the 35 cm SPC model was spent in interactive processing. This is a common feature of SPC processing: the early models require much more interactive processing than later models.

Finally, the last column of Table 2 shows the equivalent Bennu models generated later, during the actual mission operations, that correspond to each step of the process.

2.8. Quality Assessment Techniques

Normally, the quality of an SPC model is described using a measure of uncertainty because, without knowing the correct (truth) height, the error cannot be determined. However, for the

purposes of this evaluation, the synthetic asteroid provides the correct height, so we can describe the exact deviation between it and the generated DTMs. This is a true measurement of error rather than a statistical evaluation of input data to form an uncertainty. Hence, the quality of the model is assessed in two ways.

The first is via a calculation that is internal to SPC that is called the Formal Uncertainty (FormU). To calculate the FormU, SPC calculates the internal agreement of the data as a full system. Each maplet has a residual value determined by the images assigned to it. Each of these images predicts the location of the maplet, and this location is compared to the actual location of the maplet. The residual value is the rms of the difference between the predicted and actual locations of the maplet. More details will be given in E. E. Palmer et al. (2022a, in preparation). For this suite of tests, we analyzed only the highest-resolution maplets for each model. FormU is the rms of all the residual values for the highest-resolution maplets, expressed as a distance. However, the merit of the FormU has not previously been determined.

A second assessment of the quality of an SPC model is the 3D rms error. The 3D rms error requires a truth reference, so it cannot be performed during mission operations when the truth is not known. However, it provides a test of the merit of the FormU. The 3D rms value is calculated using an Iterative Closest Point (ICP) algorithm (Besl & McKay 1992; Chen & Medioni 1992), where the 3D rms is calculated as the average rms difference between every height measurement of the SPC model and the closest height of the truth model.

Two factors affect the 3D rms. The first is determined by how closely the topography follows the contours of the truth. This is the actual error that we want to identify. But the magnitude of this error can be affected by a coordinate offset. The center of an SPC model is initially defined by a priori navigation solutions. During flybys and early portions of rendezvous missions, these navigation solutions are defined by the asteroid's center of figure (CoF), especially when the gravity of an object is not measured and its center of mass (CoM) is not well defined. As a mission proceeds, and the spacecraft enters orbit, the navigation coordinate center will shift to one focused on the object's CoM. In these tests, most of the data were collected when the coordinates were not well defined relative to the CoM, but were rather in the CoF frame, which left a nonzero rms from this offset. Our ability to correctly identify the CoF–CoM offset is influenced by minor changes in the prime meridian and pole estimation uncertainties. Adjustments to the spacecraft's position and pointing during image capture are a key component of the SPC process, so there will always be some level of coordinate system shift as

Table 2
Generation of SPC Models during Preflight Testing

Process	Number of Simulated Images	GSD (cm)	Grid	Person (hr)	Computer (hr)	Equivalent Bennu Model	Equivalent Version
Ingest Approach Day 1	38	193	No	<1	<1	Limb-based draft—300 cm	
Generate 120 cm maplets	...	...	...	3	3	Mission-wide draft—150 cm	
Ingest Approach Days 2–5	152	119–175	No	7	7	Navigation draft—150 cm	
Generate 75 cm maplets	...	...	...	4	9	Navigation draft—75 cm	
Ingest Approach Days 6–9	1175	41–99	$2 \times 2, 3 \times 3$	48	46	Mission-wide draft—75 cm	v10
Ingest PS NP pass	807	12	Yes	20	21	Navigation draft—75 cm	
Ingest PS EQ pass	591	12	Yes	7	15	Navigation draft—75 cm	
Ingest PS SP pass	723	12	Yes	23	26	Navigation draft—75 cm	
Generate 35 cm maplets	...	...	...	16	138	Final—75 cm	v20
Ingest northern hemisphere BBD	1996	5	No	20	112	Navigation draft—35 cm	
Ingest southern hemisphere BBD	1969	5	No	10	101	Navigation draft—35 cm	
Generate 18 cm maplets	...	...	...	18	418	Final—35 cm	v42

Note. The processing steps required to generate the 75 and 35 cm SPC models of the synthetic asteroid, along with the number of simulated images and image GSD from the preflight testing. Person hours and computer hours give the respective amounts of time required for that processing step. The second to last column gives the equivalent model from the actual proximity operations at asteroid Bennu, and the last column gives the equivalent version of the Bennu model. Mission-wide drafts were released to the entire OSIRIS-REx team, whereas the navigation drafts were only used by the navigation team. PS: Preliminary Survey; NP: north pole; EQ: equator; SP: south pole; and BBD: Baseball Diamond.

Table 3
FormU and 3D Rms Results

SPC Model	Required Mission Residual (cm) ^a	FormU (cm) ^b	Required 3D Rms (cm)	Optimized 3D Rms (cm) ^c
75 cm	38	12	100	13
35 cm	18	6	75	10

Notes.

^a 80% of the maplets must have a residual better than this value.

^b FormU is the rms of the residuals of the highest-resolution maplets. See the text for details.

^c The optimized 3D rms is determined by removing the CoF drift before calculating the rms.

corrections are made to the spacecraft’s position, although orbital results will minimize such a bias.

A coordinate system offset leads to a systematic bias, resulting in a larger rms error than reflects the ability of SPC to duplicate a truth model accounting for a coordinate shift. The rms errors with and without the CoF drift have been determined by Barnouin et al. (2020) for the 35 cm model (their Figure 13). During operations, this CoF/CoM drift is not an issue, but with the absolute calculations done with a truth model, it needs to be removed because it is nothing but an arbitrary coordinate shift.

The ICP algorithm mentioned above can be used with or without translating and rotating the SPC model. The CoF drift is removed by translating the SPC model in several directions and calculating the 3D rms at each position, then continuing to translate in the direction that gives the lowest 3D rms. The process of following the direction of the smallest 3D rms continues until translating the SPC model in any direction produces a larger 3D rms. It is possible that this solution is a local minimum, rather than an absolute minimum, but for simple shapes, the local minimum is not an issue. This same process is used to determine the rotation that produces the smallest 3D rms. The 3D rms determined after translating and rotating the SPC model to its final position is known as the optimized 3D rms. Both the 3D rms (without translation and rotation) and optimized 3D rms (with translation and rotation) are used and discussed further in Section 3. The optimized 3D rms is then compared to the FormU. Further details of the ICP

algorithm will be found in E. E. Palmer et al. (2022a, in preparation).

Once the SPC-derived models passed the standard internal checks and the FormU requirement, processing ceased. Access to the truth model and the 3D rms (and the optimized 3D rms described in the previous paragraph) was restricted until testing was complete.

3. Results

The results for FormU and the optimized 3D rms, along with the mission test requirements, are shown in Table 3. The mission uncertainty requirement is not based upon FormU (which is the rms of the residuals of the highest-resolution maplets), but rather is a navigation requirement stating that 80% of maplets have a residual that is better than the given value. Although the mission residual requirement cannot be directly compared to the FormU, it is given as a reference since the two definitions are similar. The 75 and 35 cm SPC models have optimized 3D rms accuracies of 13 and 10 cm, respectively. As the correspondence between the truth and SPC models was high in our tests, problems of convergence of the ICP algorithm into a local minimum were not significant. The rms value is discussed in more detail below. The 35 cm SPC model offers an improvement in topographic accuracy over the 75 cm SPC model, as represented by the smaller optimized 3D rms of the former.

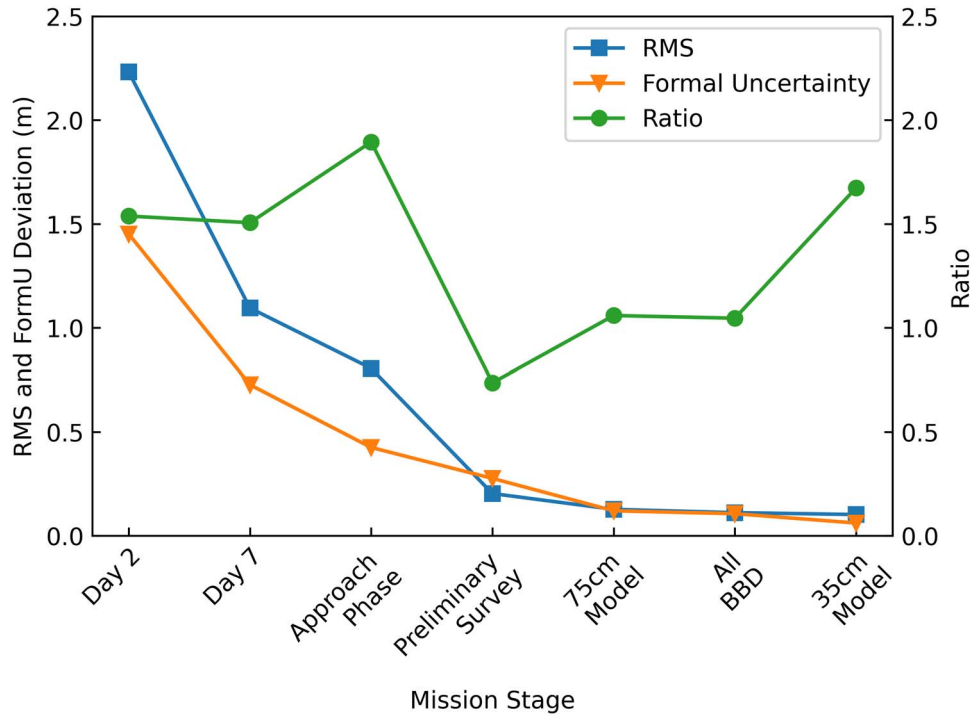


Figure 5. Progress of optimized 3D rms deviation (left y-axis), FormU (left y-axis), and the ratio between them (right y-axis), through the simulated mission stages. Optimized 3D rms and FormU follow each other closely and are always within a factor of two. This shows that FormU (which can be measured in flight) is a close proxy for the optimized 3D rms error (which cannot be measured in flight). The tests show that overtilting (making higher-GSD maplets than the final DTM) improves the rms error of the model. We note a 37% improvement in rms for the “75 cm Model” when we create 35 cm GSD maplets with no additional data, and a further 8% improvement when we create 18 cm maplets for the 35 cm GSD model.

The accuracy of the SPC-determined spacecraft position and pointing is not a direct indication of the accuracy of the SPC model. Still, it is worth noting that the rms of all the position errors of each mission phase (Approach, Preliminary Survey, etc.) are less than the 1σ flight errors of the kernels (see Section 2.3). For example, the spacecraft position error during the simulated Detailed Survey was 4.3 m, compared to the kernel error of 6.4 m.

Figure 4 (top) shows a simple cylindrical projection of the radius of the truth model. Figure 4 (middle) shows the radius difference between the generated 35 cm SPC model and the truth model, while Figure 4 (bottom) shows the radius difference as a histogram. This figure indicates that the sectors all have consistent radius differences, showing that different crater and boulder density and roughness factors are all accurately represented by SPC. The greatest error of the 35 cm SPC model is 1.5 m, but it is typically less. Figure 4 (bottom) shows there is a slight skew to positive values.

Most of the error occurs at the poles, but other errors occur in small patches on the model. The errors at the pole are because of shadows and uncertainties in the albedo estimates provided by SPC. Images with low incidence angles, which could not be acquired owing to Bennu’s low axial tilt, would have illuminated all portions of the surface and constrained the albedo. Without tight constraints on the albedo, the radius of each vertex near the pole has higher error. This higher error occurs because SPC cannot fully deconvolve slope from albedo, and is discussed further by Palmer et al. (2022). The errors occurring equatorward of 45° latitude have small lateral extent. These deviations do not occur near large boulders or changes in albedo, but instead occur in topographically subdued (synthetic) areas, and the images are correctly aligned

in these locations. Given the small lateral extent, the contribution to the overall error is small.

The optimized 3D rms (which we cannot measure in flight) and the FormU (which we can measure in flight) are closely correlated, as seen in Figure 5. This is an exceptional result for an SPC model whose radius is only represented laterally every 70 cm. Although these two metrics are not equivalent, they remain within a factor of two of one another for each stage of the SPC model development, as can be seen by the ratio in Figure 5. See Section 4 for further discussion of the ratio.

Although the global tests did not continue to finer resolutions than the 35 cm topography, the simulated images were still ideal for building robust topography at 12 cm. The simulated Baseball Diamond images help to round out the ideal imaging parameters at 12 cm, but they are only sufficient, not ideal, at 5 cm, even when combined with the simulated Preliminary Survey images (see E. E. Palmer et al. 2022b, in preparation). We generated 8 cm topography at the simulated sampling site to demonstrate the high quality of the model created. Figure 6 (top) shows the height difference between the truth model and the SPC model, which is typically ≤ 5 cm. Figure 6 (bottom) shows the height difference as a histogram, which has a slight skew to positive values. The largest craters shown here are rendered about 10 cm too shallow (yellow) by SPC, and the largest boulders are about 10 cm too short (dark blue). The maximum deviation is around 15 cm, which is close to the ideal image set of 12 cm pixel^{-1} (i.e., that of the simulated Preliminary Survey images).

The typical and maximum errors of the 8 cm topography allow us to make some general statements about the errors associated with an ideal and sufficient image set. In these tests, Preliminary Survey and Baseball Diamond images combine to

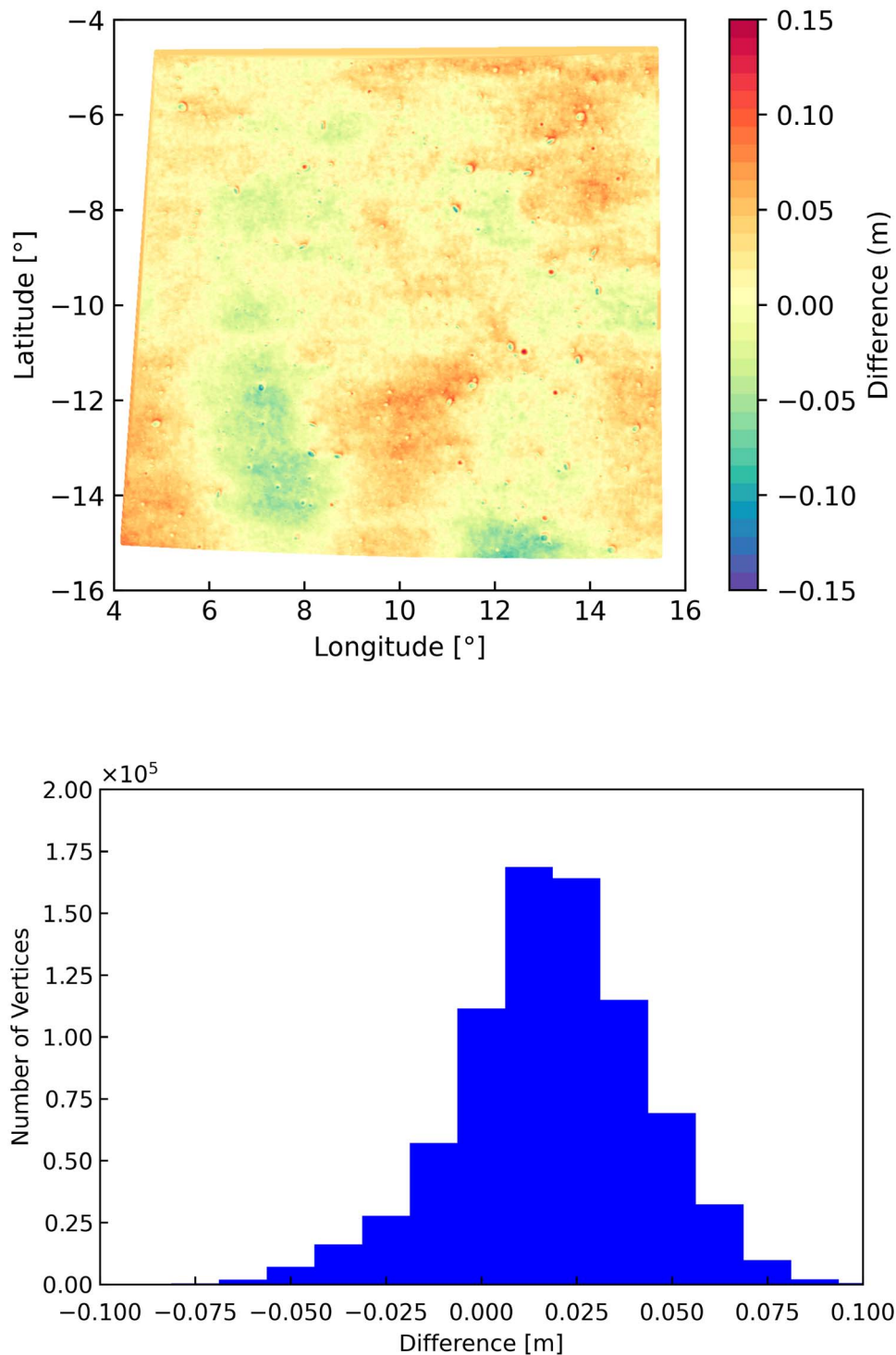


Figure 6. Top: the height difference between the truth model and local DTM of an arbitrary sampling site as a simple cylindrical projection. Topography is represented every 8 cm. The simulated image suite is ideal at 12 cm and sufficient at 5 cm. Bottom: the height difference from the top panel displayed as a histogram.

create an ideal image set at 12 cm pixel^{-1} and a sufficient image set at 5 cm pixel^{-1} . Since the maximum deviation of the topography is 15 cm, which is close to the 12 cm pixel^{-1} resolution of the ideal image set, we conclude that the error of the topography under ideal conditions is, at worst, about 1 image pixel, and typically much less. Since the typical deviation of the topography is $\leq 5 \text{ cm}$, which is the same as the 5 cm pixel^{-1} resolution of the sufficient image set, we conclude that under sufficient, but not ideal, conditions, the error is

typically 1 pixel of the best image. The maximum deviation of the topography indicates that the error is at worst 3 image pixels of the best image.

4. Discussion

In addition to establishing SPC as class B software for NASA and understanding the quality of the global DTM produced by SPC, the testing presented here was critical for refining the imaging plan flown during the OSIRIS-REx

proximity operations. These preflight tests showed that a 75 cm SPC model did not require PolyCam images to be collected during Preliminary Survey, but could instead use MapCam images. This was important when the mission discovered that the data storage capacity of the spacecraft would be exceeded if using PolyCam for Preliminary Survey. Lower-resolution MapCam images were enough to build the 75 cm DTM and meet the navigation requirements, and far fewer images were required to obtain complete coverage of Bennu. We also discovered that the Baseball Diamond stations in these tests, while sufficient for a 35 cm global DTM, could be greatly improved upon for both stereo and SPC to support the navigation needs for higher-resolution regional DTMs. The regional DTMs are not limited by the 1.6 million vectors of a global DTM, and can benefit from higher-resolution images. This discovery provided the impetus to increase the number of viewing stations for the Baseball Diamond campaign. Although the simulated campaign had four Baseball Diamond flybys, each flyby covered only half of the truth model, resulting in four stations for each maplet. For the as-flown mission, the number of Baseball Diamond stations for each maplet was increased from four to seven (Lauretta et al. 2021).

These tests have demonstrated the effects of generating high-GSD maplets on the optimized 3D rms. Although the SPC-derived global DTM represents the radius every 70 cm, the optimized 3D rms improves by 37% after generating the 35 cm maplets, and 8% after generating the 18 cm maplets (Figure 5). This improvement is due to better alignment of the images, and shows that increasing the resolution of the maplets beyond the GSD of the final DTM can have a significant effect on the accuracy of the DTM.

Figure 5 demonstrates that the FormU provides a good quantitative estimate of the model fidelity over the entire mission. In all cases, the correspondence is within a factor of two. Early in the mission, when the image suite is not as robust, the FormU slightly overestimates the fidelity of the model. After a robust image set is obtained and fully processed, the FormU provides an excellent estimate of the model fidelity. Some small growth in the difference between the FormU and the rms for the 35 cm SPC model exists. This difference may be the result of the optimized 3D rms converging to a minimum (perhaps because the global shape is limited to a GSD of 70 cm), while the FormU is based upon the maplets that are only limited by the image resolution of 5 cm GSD. From the authors' experiences with SPC, our preferred interpretation for the variation of the ratio is that it represents noise. Additional testing would be required to identify if variations in the ratio are due to data quality, amount of processing, or other factors.

Finally, this testing showed that the preflight procedures and scripts for the mission were generally adequate. We verified which maplet resolutions to use, the number of iterations, and the number of needed updates to the spacecraft position and pointing. There were, however, three main differences in processing between these tests and the proximity operations. The first was using limb information from early Approach images to refine the Nolan et al. (2013a) shape in flight. This made the processing of SPC Day 1 images smoother. The second was additional interactive processing on large boulders. Standard processing resulted in underestimated boulder heights, so during the proximity operations, we placed additional maplets on the tops of boulders to ensure that a stereo point would pull the correct height. Third, we performed

additional work during the proximity operations to obtain an improved pole position, which was not part of these tests.

5. Conclusion

We used a computer-generated synthetic ("truth") model of asteroid Bennu to simulate the images that would be acquired during the OSIRIS-REx mission and to test the efficacy of SPC-based shape modeling. In particular, per the mission's requirements for science and navigation, we used the simulated images to generate a 75 cm SPC model from 75 cm GSD maplets and 12 cm pixel⁻¹ images, and a 35 cm SPC model from 35 cm GSD maplets and 5 cm pixel⁻¹ images. The success of this testing, which took place before the spacecraft encounter, defined the procedures used for SPC-driven imaging and DTM generation in flight.

This study shows that with carefully planned observations, the accuracy of both of the SPC-derived global DTMs is better than the 70 cm lateral representation of the radius. By using the ICP algorithm to remove the CoF drift introduced by SPC processing, we find the optimized 3D rms of the 75 cm SPC model to be 13 cm, and that of the 35 cm model to be 10 cm. These results, along with other testing (E. E. Palmer et al. 2022b, in preparation), in confirm that the accuracy of the DTM is about the same as the pixel size of the images when there is a sufficient image set. Although the CoF drift of the SPC model can be removed in flight, the 3D rms accuracy cannot be determined in flight.

SPC's Formal Uncertainty (FormU), a measurement of the internal consistency of the model that can be performed in flight, for the 75 cm SPC model is 12 cm, and for the 35 cm SPC model is 10 cm. More importantly, we find that the FormU is always within a factor of two of the optimized 3D rms. Thus, the FormU is a good indicator of the uncertainty of the topography. With the FormU, we can determine more than just the uncertainty of the Bennu SPC model; we can also determine the FormU (and hence the uncertainty) for future global DTMs, as well as for global DTMs already generated.

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All data from the OSIRIS-REx mission are available via the Planetary Data System at <https://sbn.psi.edu/pds/resource/orex/>. The global DTMs of Bennu produced from radar and flight data are available via the Small Body Mapping Tool at <http://sbmt.jhuapl.edu/>. Licenses for the SPC software suite are available from the Planetary Science Institute at <https://spc.psi.edu>. The test data and SPC models for this work can be found at <https://doi.org/10.26033/v0s3-z379> (Palmer & Gaskell 2021).

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